



Temperature

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1. What *Temperature* Means

Temperature is a hyper-parameter that shapes **how random** the language model's token sampling will be.

- **Low temperature** → **deterministic** (model gravitates toward the highest-probability token).
- **High temperature** → **stochastic** (model explores lower-probability tokens, adding variety).

2. How It Works Under the Hood

1. The model assigns a probability to every possible next token.
2. Those probabilities are divided by the temperature value T and then renormalized.
3. A token is sampled from this adjusted distribution.
 - $T < 1$: Distribution becomes *sharper* → top tokens dominate.
 - $T > 1$: Distribution becomes *flatter* → tail tokens gain weight.

3. Typical Temperature Bands

Band	Output Character	When to Use
0 – 0.3	Almost deterministic, highly factual, reproducible	Math, factual Q&A, code generation, policy compliance
0.5 – 0.7	Balanced creativity and coherence	Most general-purpose chat, product descriptions, summaries
0.8 – 1.0	Highly creative, novel, more risk of drift or hallucination	Brainstorming, fiction, poetry, marketing slogans

Rule of thumb: Lower T for precision, higher T for imagination.

4. Choosing the Right Temperature

4.1 Tasks Requiring Precision

- Arithmetic, step-by-step reasoning, legal clauses
- Recommended **T = 0.0 – 0.2**
- Pair with **top-p ≤ 0.3** and **max_tokens** set just large enough for the answer.

4.2 Balanced Informational Content

- Technical explanations, tutorials, conversational agents
- Recommended **T = 0.4 – 0.6**
- Keep **top-p ≈ 0.9** for smoother phrasing while limiting tangents.

4.3 Open-Ended Creativity

- Storytelling, songwriting, idea generation
- Recommended **T = 0.8 – 1.0** (sometimes higher if the API allows)
- May combine with **top-p = 1.0** for maximal diversity, but be prepared to post-edit.

5. Best-Practice Tips

1. **Start low, scale up:** Test with **T = 0** and increase gradually until outputs feel “alive” but still on-topic.
2. **Use temperature + top-p together:** Temperature controls global randomness; top-p trims long tails.
3. **Batch sampling for ideation:** Generate multiple completions at high T, then curate manually.
4. **Keep prompt quality high:** A weak prompt plus low T can still yield bland text; clarify instructions first.
5. **Monitor hallucinations:** As T rises, so does the likelihood of fabricated facts—fact-check accordingly.

6. Common Pitfalls

- **Over-indexing on T alone:** Neglecting stop sequences, max_tokens, or system instructions can override temperature effects.
- **Assuming high T always equals better creativity:** At very high values (>1.2) coherence can break down rapidly.
- **One-size-fits-all mindset:** Optimal temperature varies by *both* domain and user tolerance for novelty.

7. Quick Decision Flow

1. **Does accuracy trump novelty?** → *Yes* → Start with **T = 0–0.2**.
2. **Do you need a mix of correctness and tone?** → *Yes* → Try **T = 0.4–0.6**.
3. **Is the goal divergent thinking or inspiration?** → *Yes* → Use **T = 0.8+** and sample multiple outputs.

Key takeaway: Temperature is your *creativity dial*. Turn it down for reliability, turn it up for originality—always validating the result against the task’s success criteria.